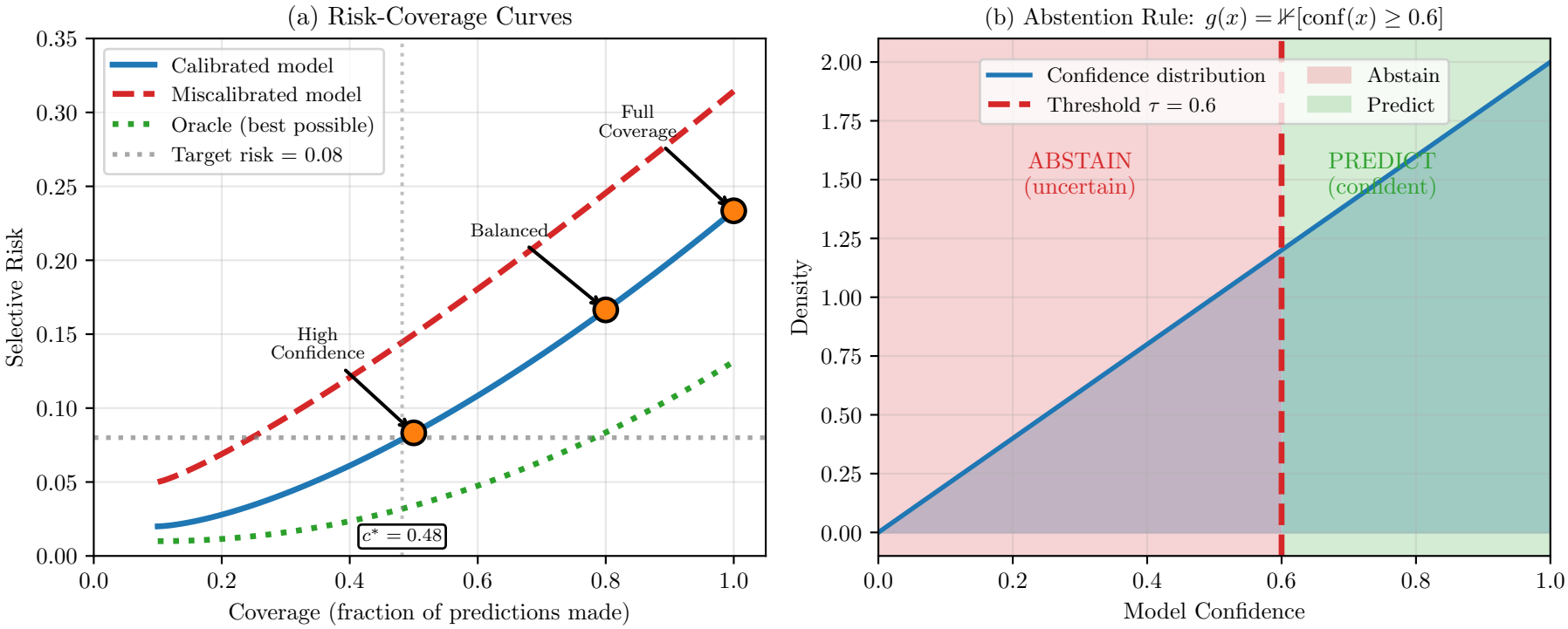
(a) Risk-Coverage Curves
Calibrated model
Miscalibrated model
Oracle (best possible)
Target risk = 0.08
Full Coverage
Balanced
High Confidence
$c^* = 0.48$
Selective Risk
Coverage (fraction of predictions made)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.0
0.2
0.4
0.6
0.8
1.0
(b) Abstention Rule: $g(x) = \mathbb{1}[\text{conf}(x) \geq 0.6]$
Confidence distribution
Threshold $\tau = 0.6$
Abstain
Predict
ABSTAIN (uncertain)
PREDICT (confident)
Density
Model Confidence
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0.0
0.2
0.4
0.6
0.8
1.0